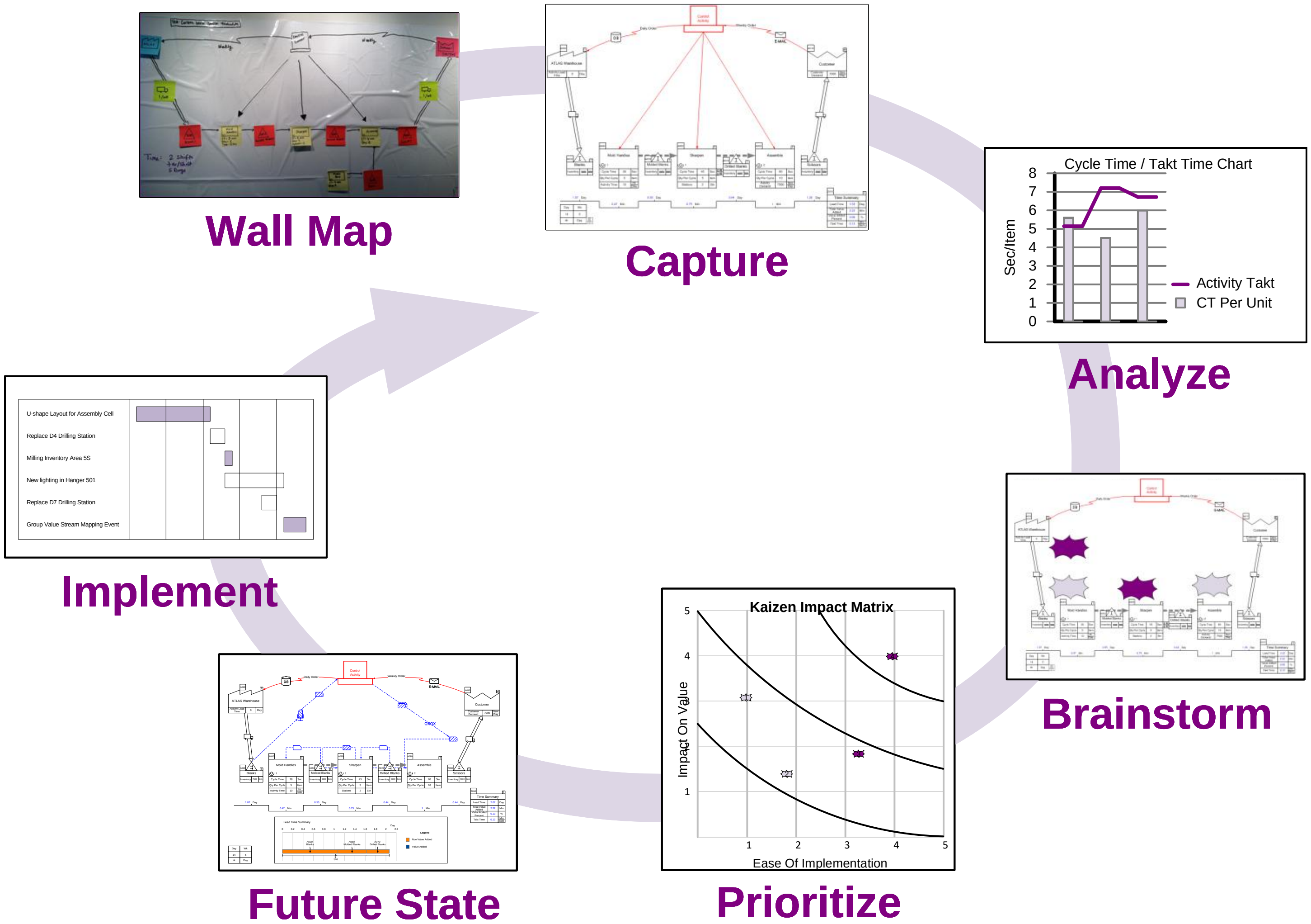
Lean VSM Intro

Overview of Lean concepts and value stream mapping



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How to Use this File

This file contains the reading materials and the exercise pages from the course (title on previous page). While the course can only be taken on a computer, this booklet can be useful for note taking and later for refresher training.

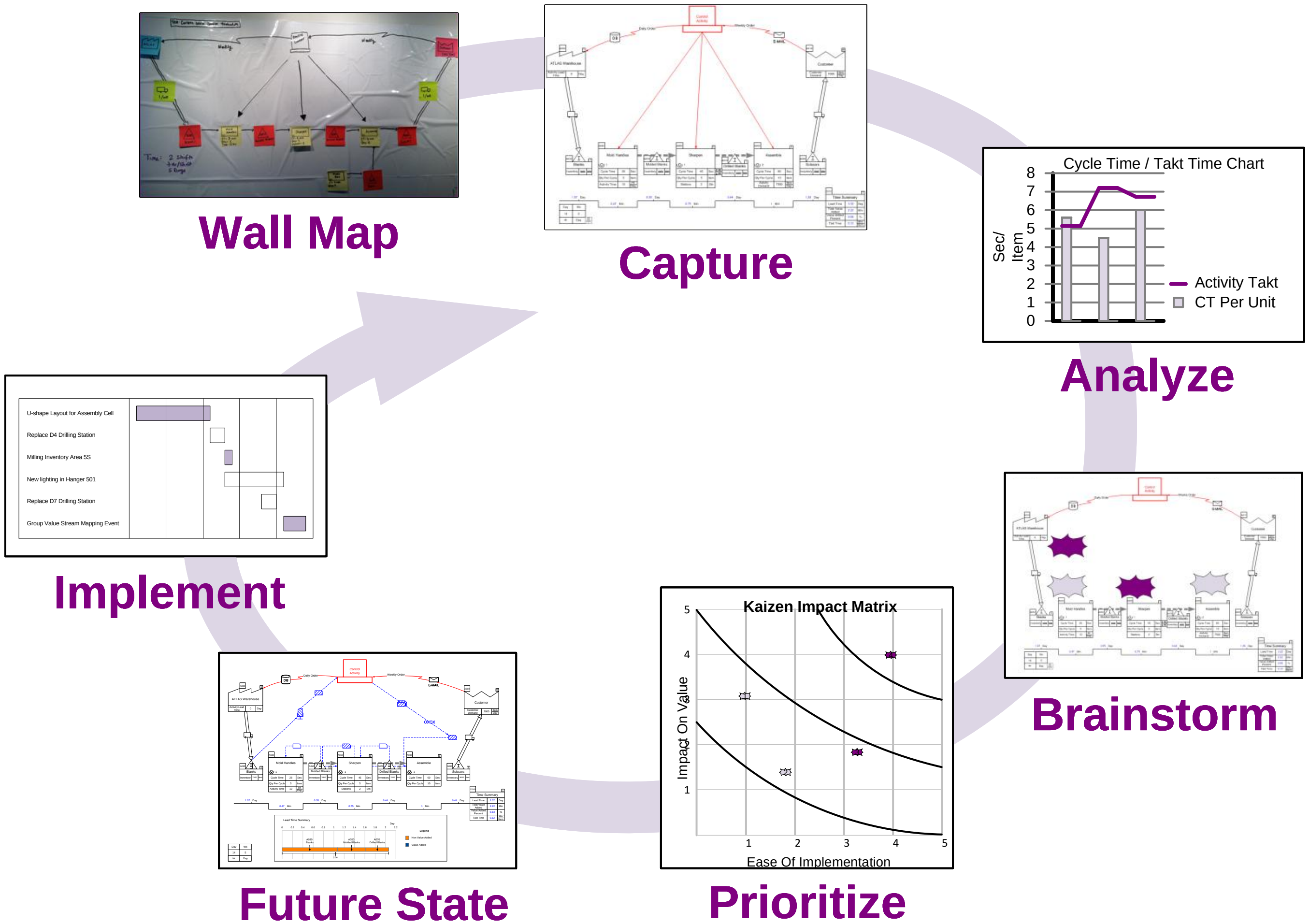
This booklet is designed for on screen and print use. For on screen use, we recommend Acrobat Reader with the page display set to "Single Page View". If you are using this booklet on-screen while going through the exercises in eVSM, a second monitor is very helpful.

For hardcopy use, print the file on 8.5x11 or A4, and bind along the long edge.

Table of Contents

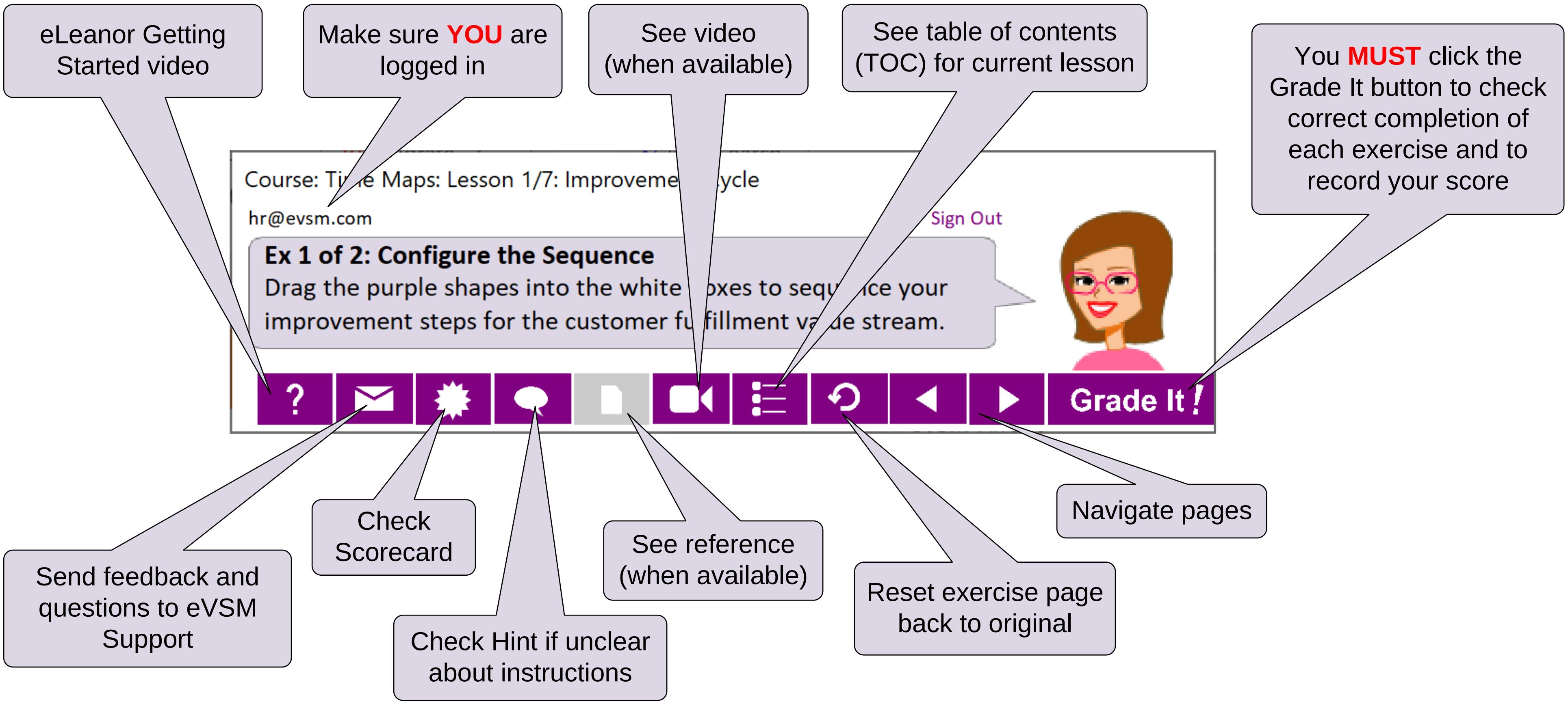
Improvement Cycle with Value Stream Maps

Focusing efforts on key value streams and removing the waste in them is central to Lean thinking. In this lesson, we will study the components of a typical map and look at the improvement cycle within which such maps are used.



Improvement Cycle with Value Stream Maps

Working with the eLeanor Control Panel

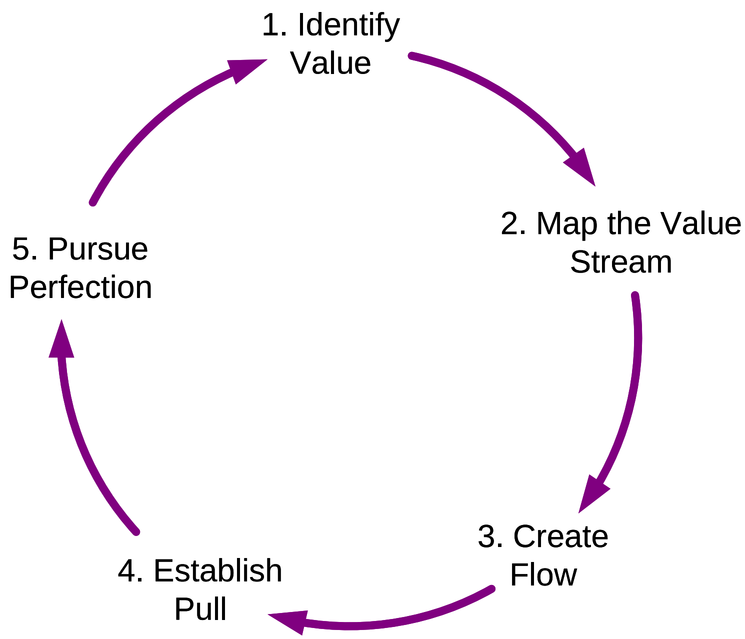


Important Notes

1. Make sure you have a good eLeanor environment: large screen PC, 1280x720 resolution minimum, physical mouse with scroll wheel
2. When you complete an exercise, you MUST click the “Grade It” button
3. You WILL lose points if you get an exercise wrong the first time
4. If you are stuck on an exercise, check the Hint. If that does not help, go back and review the preceding Readme pages. If you are still unsure, click the Feedback button in the eLeanor panel and ask your question.

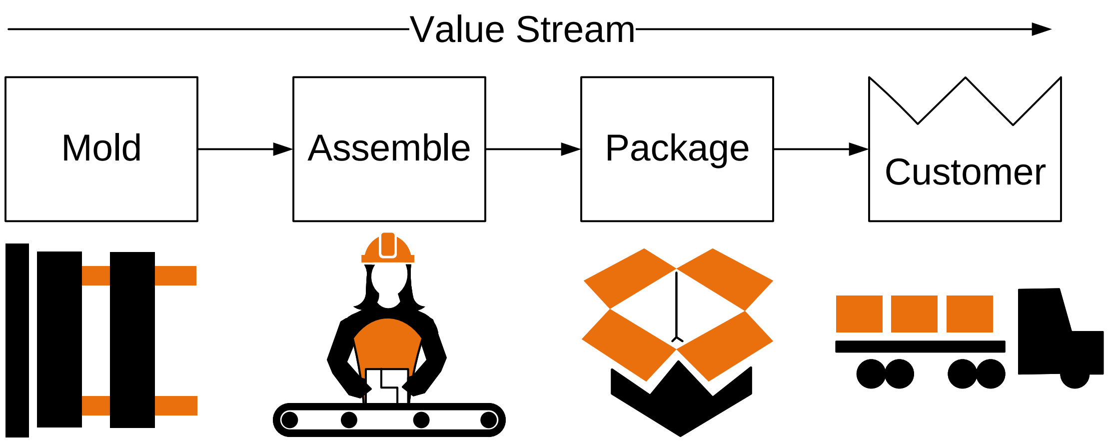
Lean Principles

The book “Lean Thinking” by Womack and Jones introduces the principles below of Lean Thinking. It starts with selection of key value streams which are then improved following these principles. Note the early “Map the Value Stream” step.



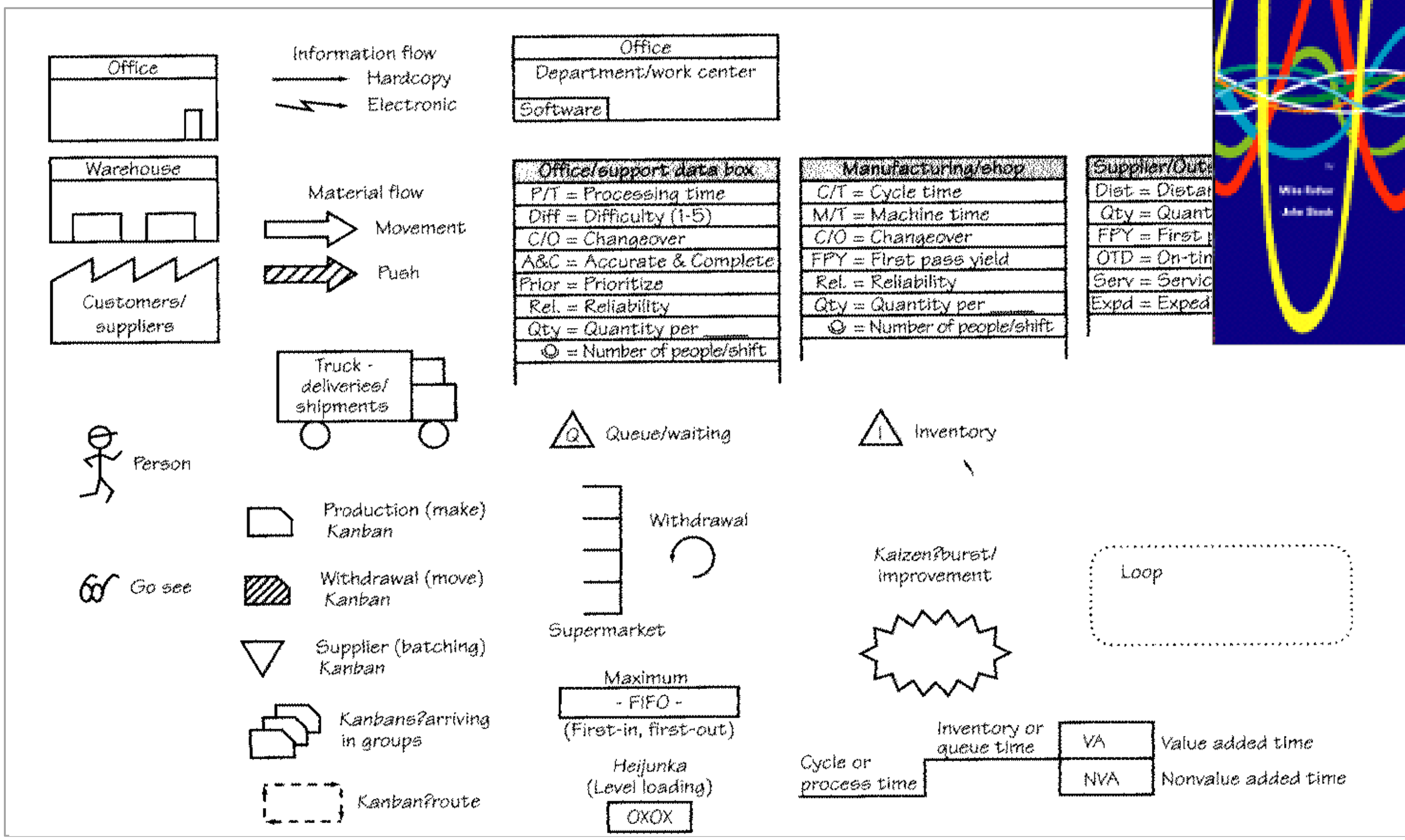
What is a Value Stream?

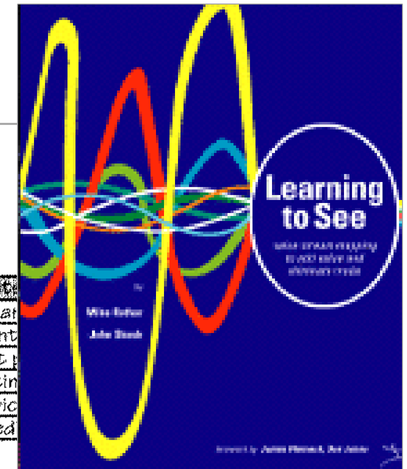
A value stream is all of the steps, both Value Added and Non Value Added that are required to complete a product or service from beginning to end. Customers could be external or internal to an organization.



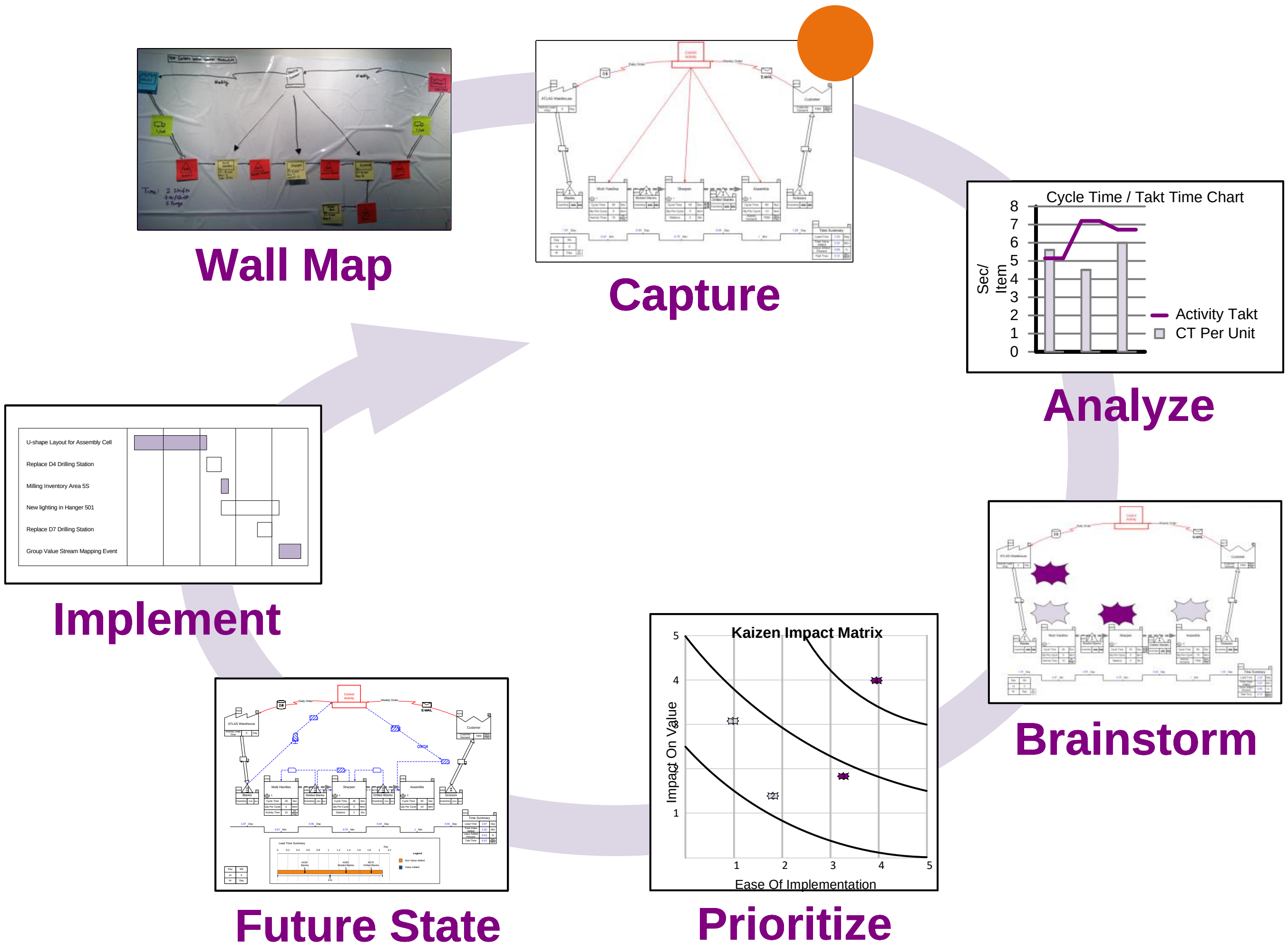
Origins of Value Stream Mapping

The book "Learning to See" introduced the term "value stream mapping" and remains a great reference on the subject. Below are icons and data blocks it recommends for constructing these maps.



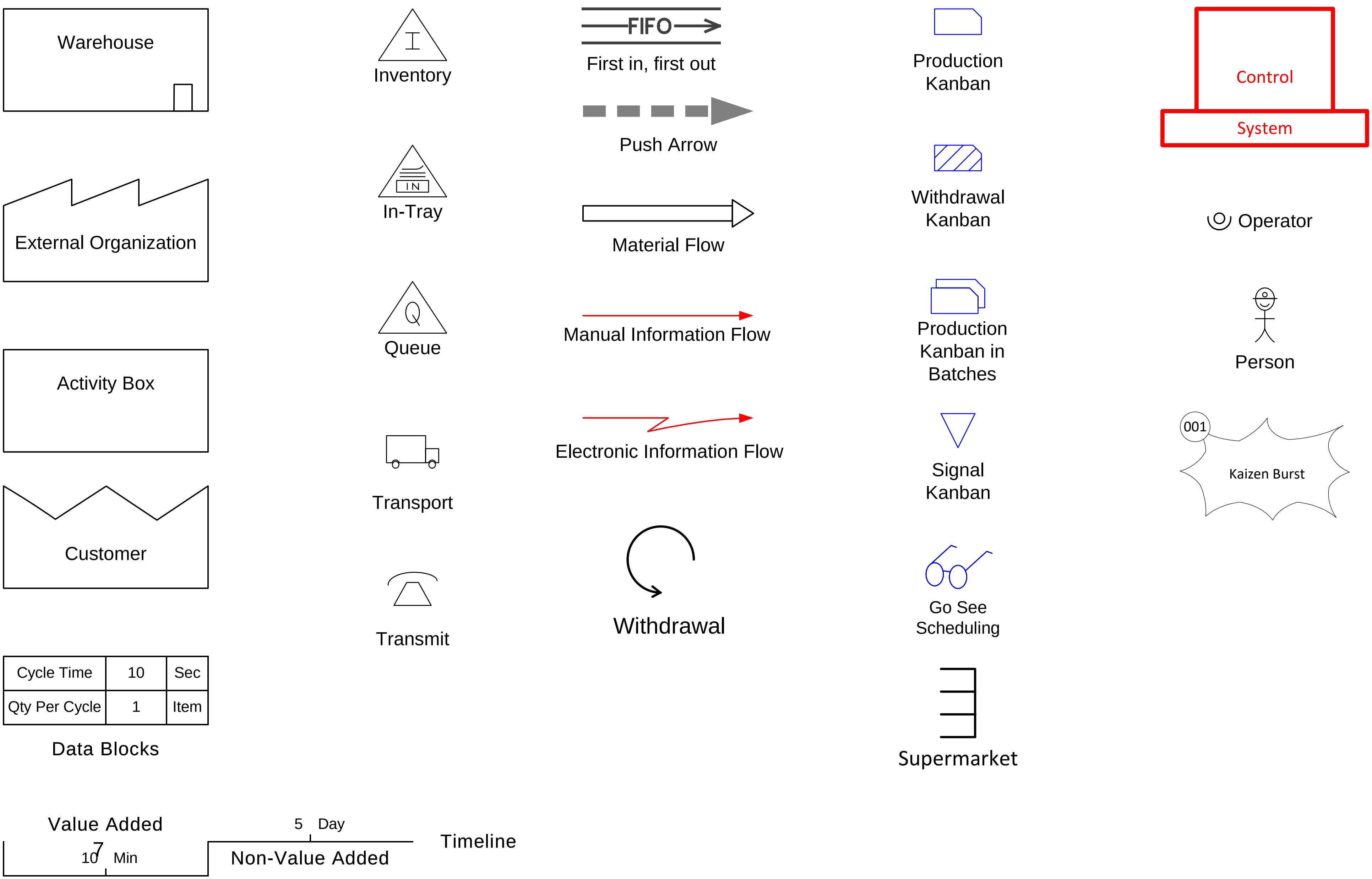


Capturing the Map



Value Stream Mapping Icons in eVSM

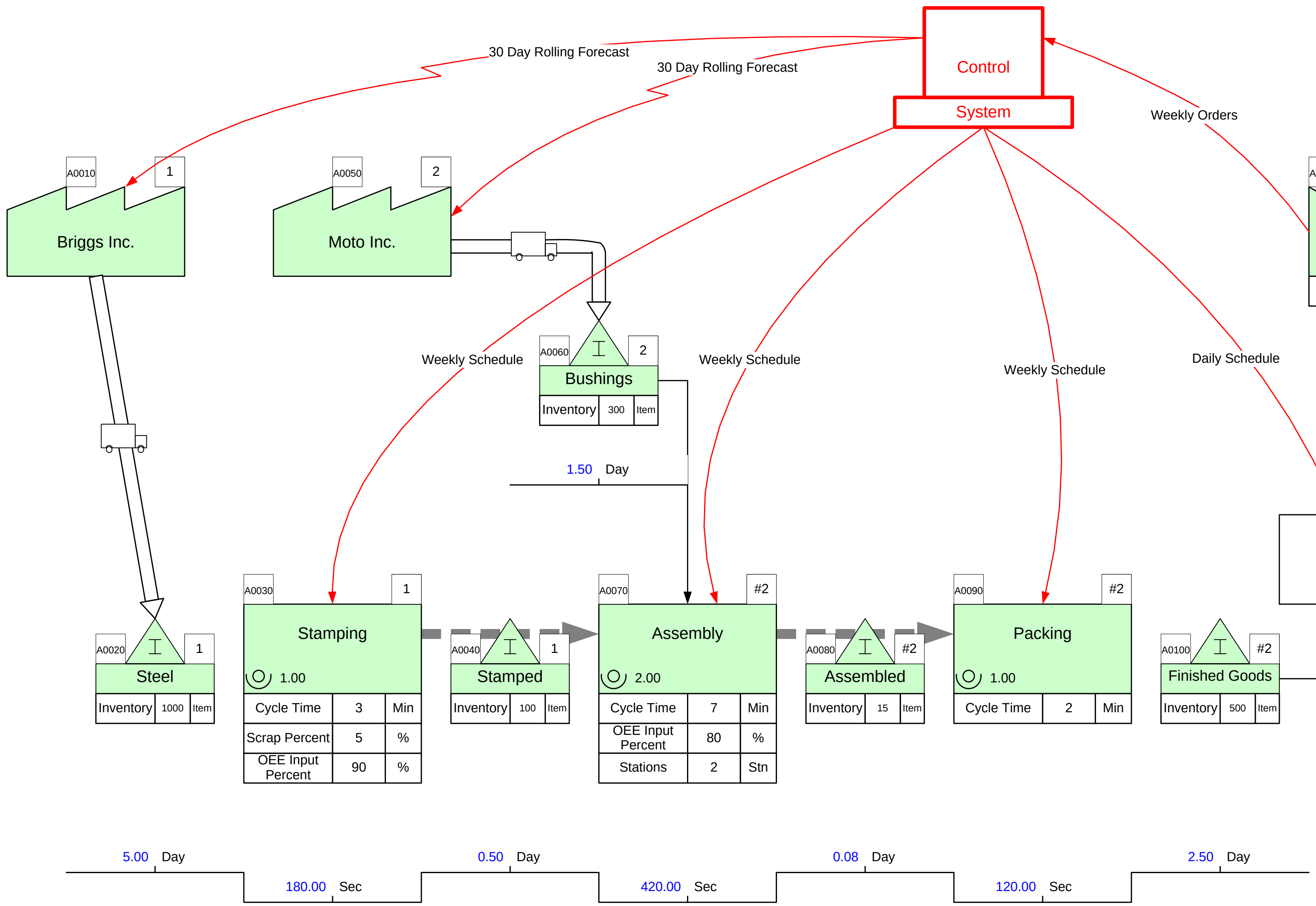
eVSM has electronic “drag and drop” versions of the original icons from “Learning to See” but expanded to provide a complete set to quickly draw a map.



Information Flow and Material Flow

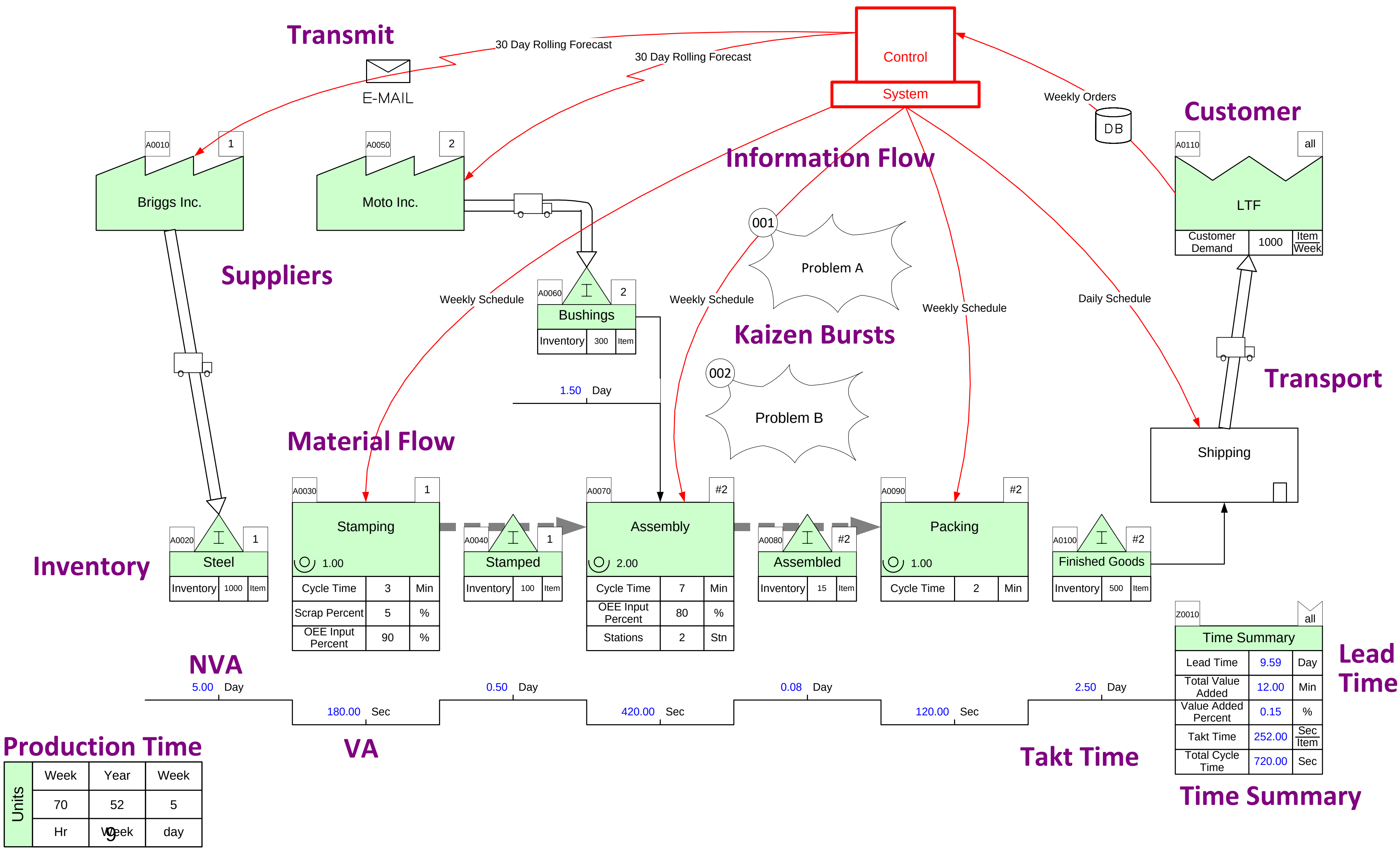
Here's an example value stream map drawn in eVSM.

The "Control" center and red arrows represent the information flow and guides what to make and when to make it. The rest of the map is the material flow of the production process. At the very bottom is the production timeline.



Value Stream Map Components

This shows the components within the overall material and information flows.



Q. Which ONE of these is typically NOT a component of a plant level value stream map?

◎ The material flow along the production process.

◎ The information flow that guides what to make and when to make it.

◎ Kaizen bursts that show current problems and potential improvements in the value stream.

◎ Planned production each week in the next month or 3 months.

◎ Customer(s) and customer demand.

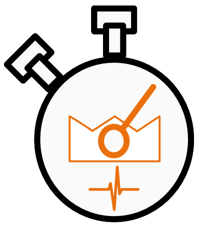
◎ A VA/NVA timeline that helps understand the overall lead time.

◎ Plant production time (actual work time) information.

Takt Time

Synchronizes pace of production to match pace of customer demand. By synchronizing production to demand we reduce the worst type of waste in the value stream, overproduction.

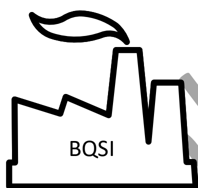
$$\frac{\text{Available Time}}{\text{Demand}}$$



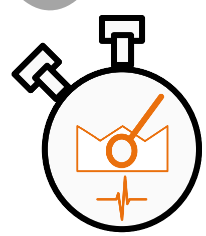
$$\text{Takt Time} = \frac{\text{Effective Working Time per Period}}{\text{Customer Requirement per Period}}$$

Calculate the takt time in the example below.

- ◎ 6 Min / Item
- ◎ 8 Min / Item
- ◎ 10 Min / Item
- ◎ 12 Min / Item



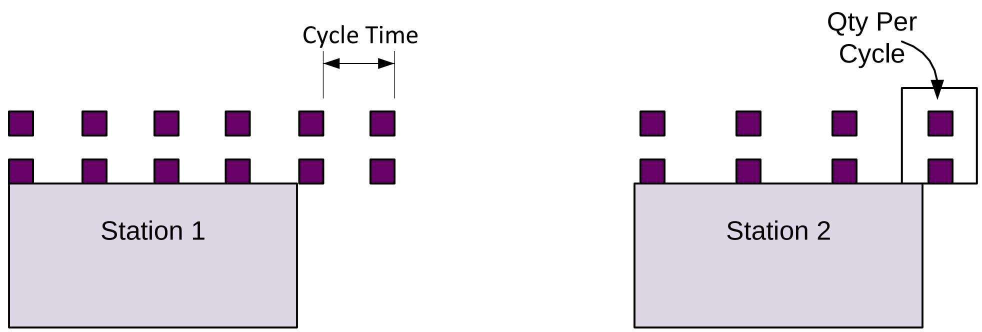
70 Production Hours / Week



700 Items / Week

Activity Time Related Terms

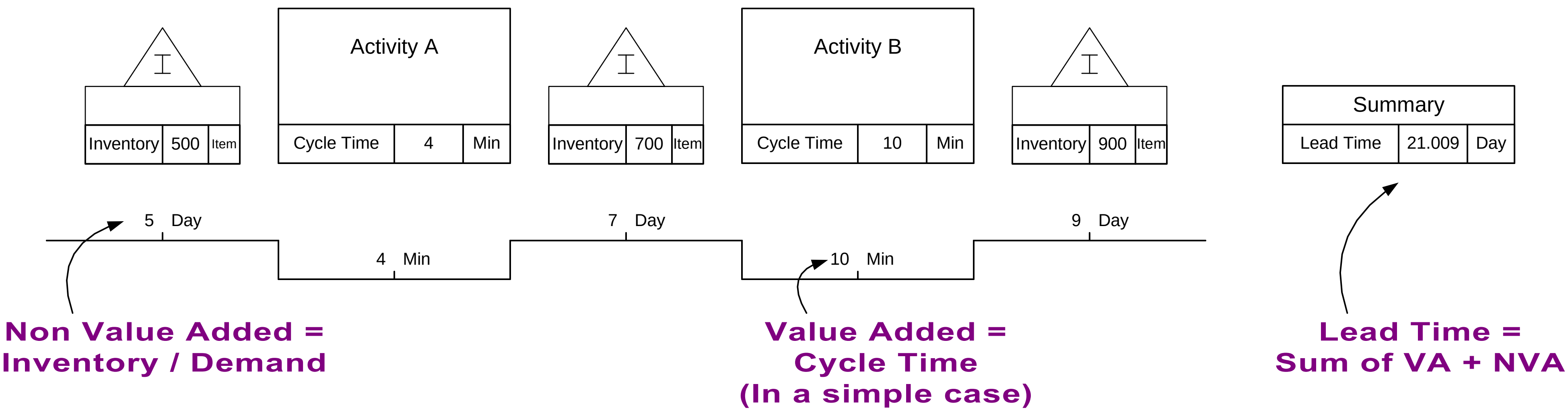
- Cycle Time: The time interval between a set of items exiting an activity and the next set.
- Qty Per Cycle: The number of items produced during one cycle at one station.



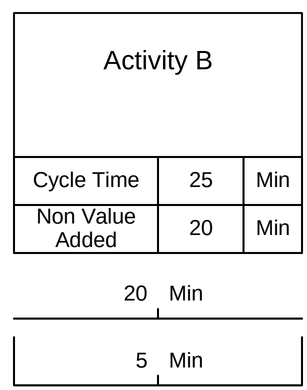
Lead Time, VA, and NVA

We can draw a timeline under the activity boxes and inventories to help us compile the production lead time. The “valleys” in the timeline correspond to Value Added time. The “hills” correspond to Non Value Added time.

Value added time for an activity box is the time that value-added work is being done on an item moving through the process. In the simplest case value added time will equal cycle time.



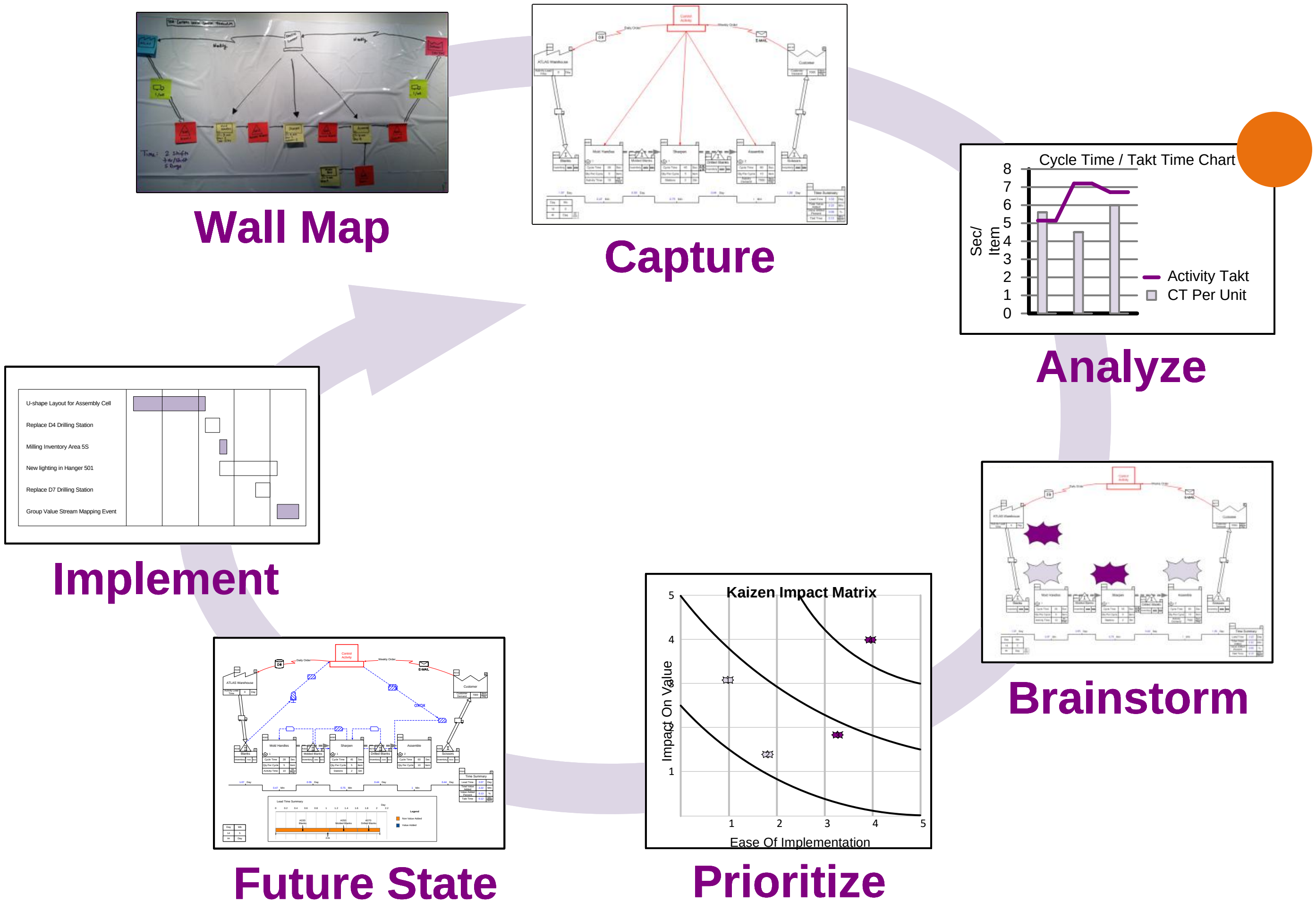
As a variation, an item could be in an activity for 25 mins (lead time) of which 5 minutes may be value added time and the rest non-value added. This would be shown with 5 mins in a “valley” and 20 mins in a “hill” under the activity.



Units		
	Day	Hr
	24	60
	Hr	Min

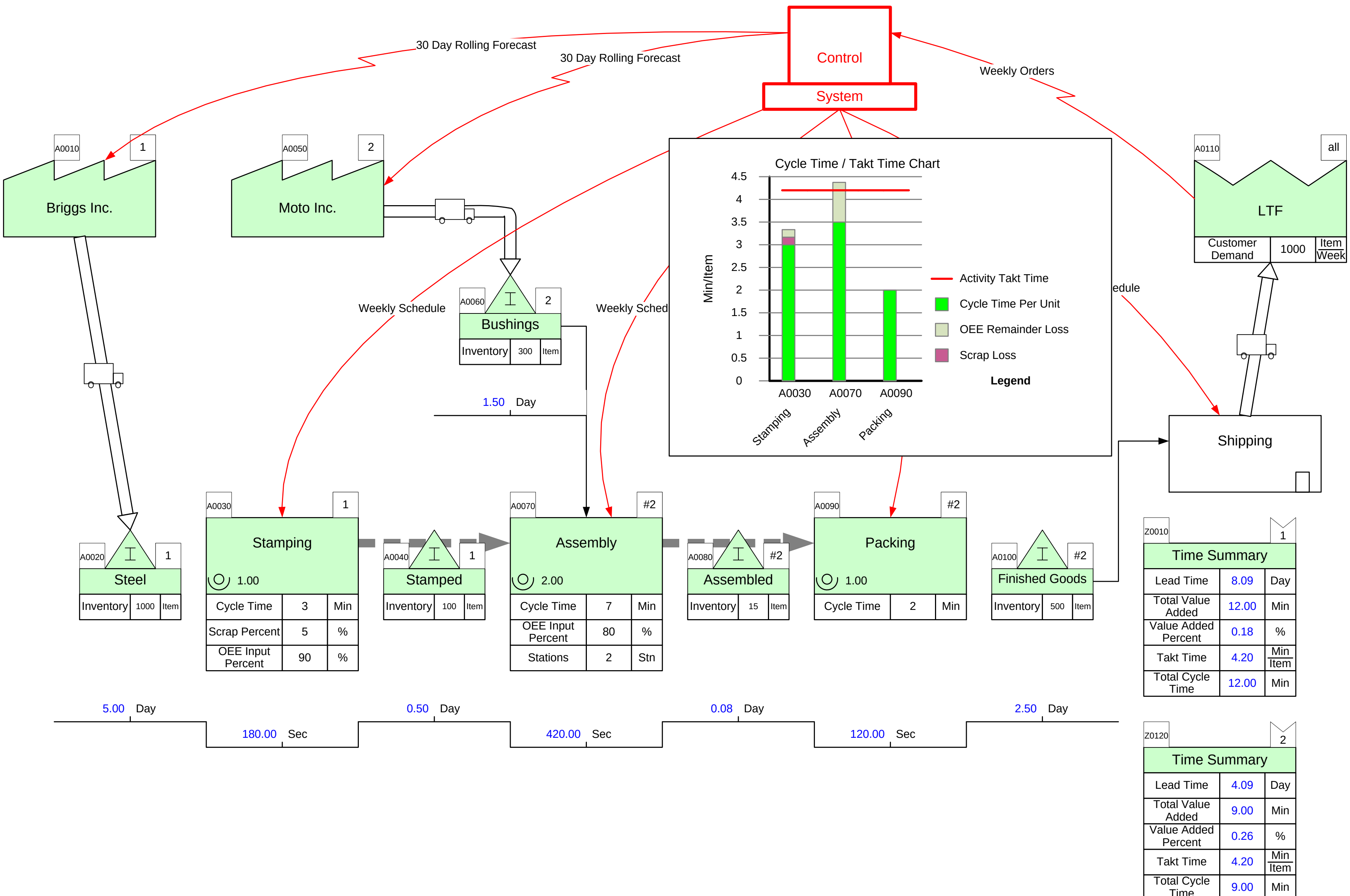
Analyzing the Map

We can analyze the map for sources of waste and their magnitudes. Visuals of production Lead Time and Capacity support this process.



Capacity Visualization

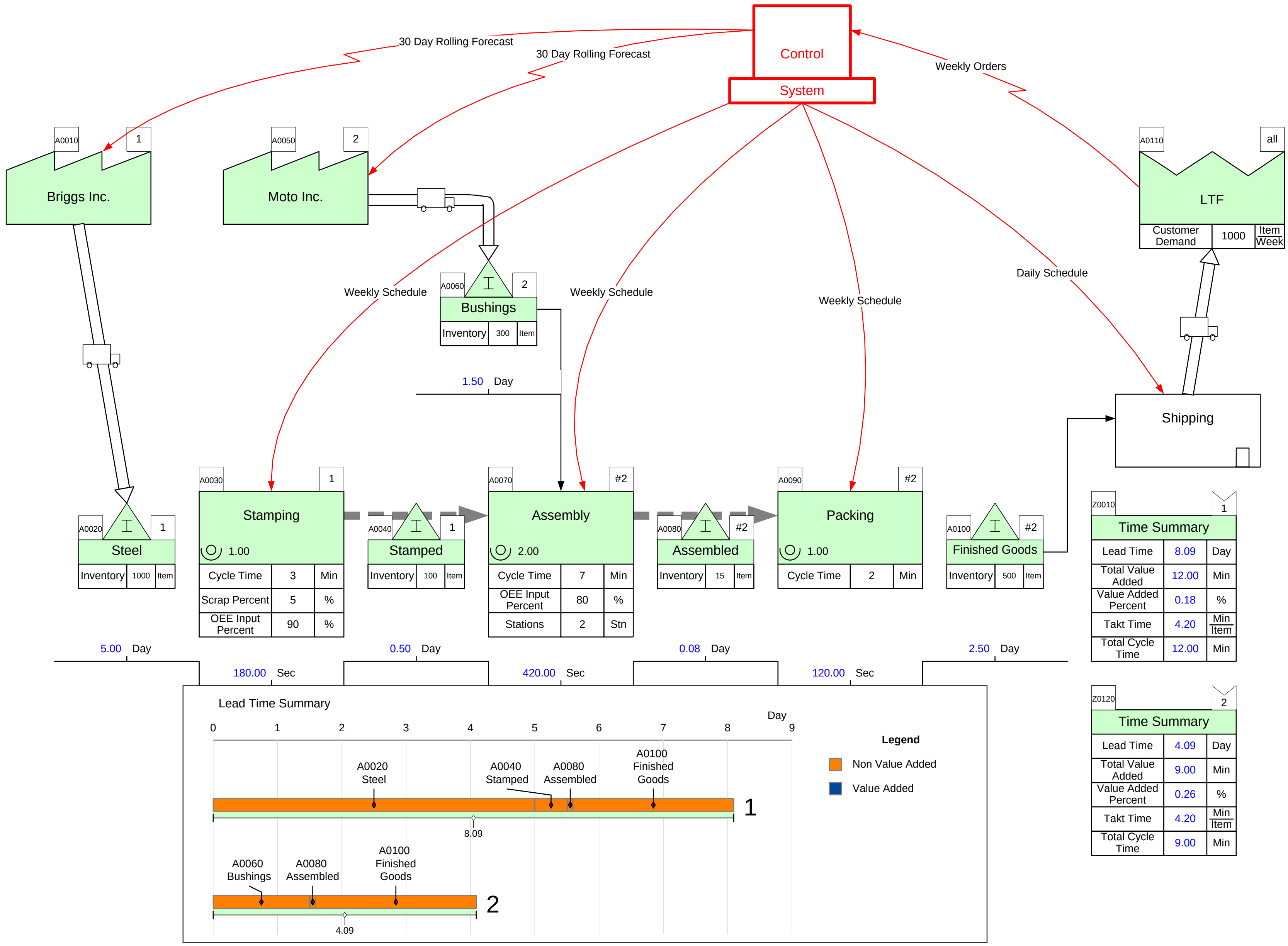
The Cycle Time / Takt Time chart compares what the customer needs to what we are able to make at each station.



Lead Time Visualization

For each path through the value stream, the Lead Time Chart shows the VA/NVA time components. It's really a "to scale" view of the VA/NVA timeline drawn at the bottom of most maps.

Note that the VA times are so small compared to the NVA times that they are not "visible" on the lead time chart.



Q. Which of the following statements about Lead Time are true? Tick all 4 that are TRUE.

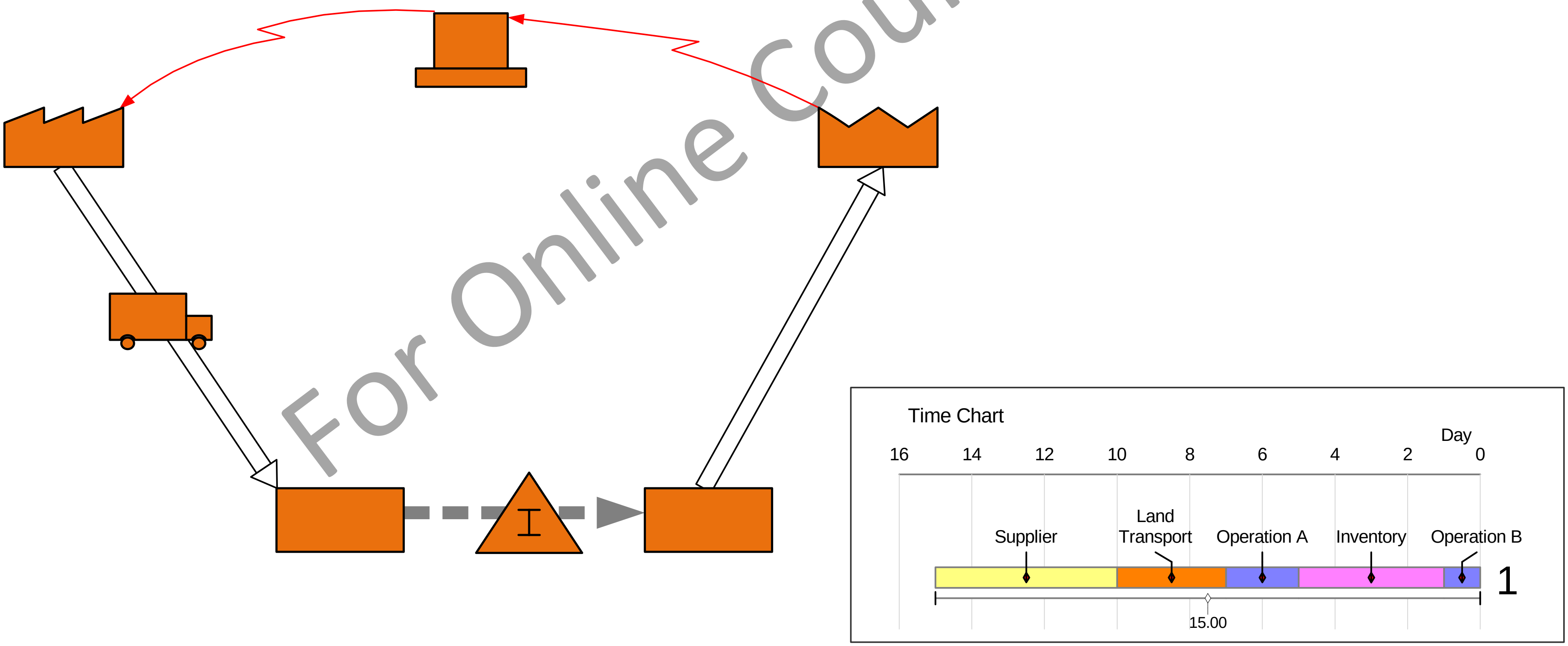
☐ Lead Time includes Transport Time

☐ Lead Time includes waiting in inventory

☐ Lead Time includes “Value Added” time

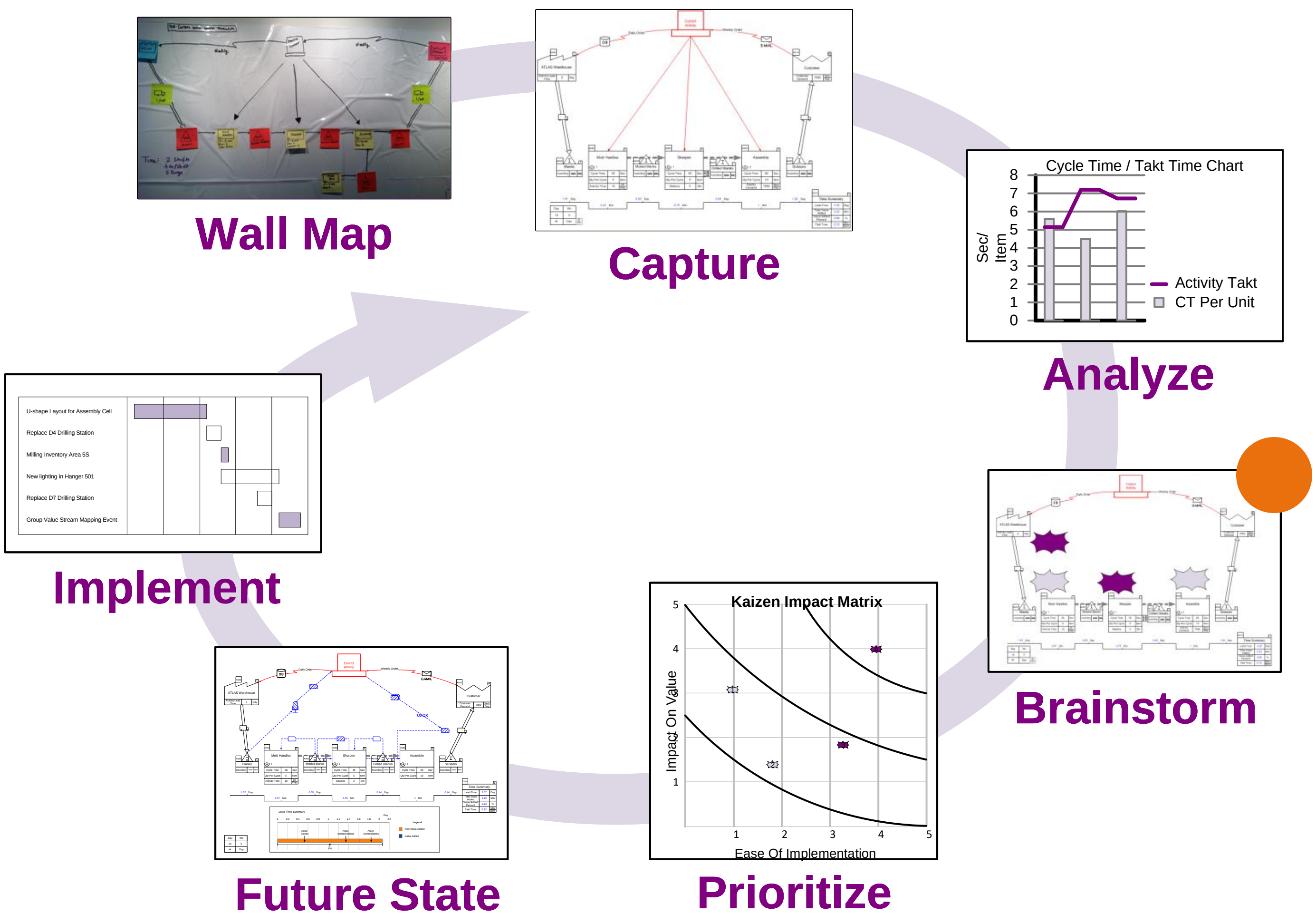
☐ Long Lead Times are always due to inadequate capacity

☐ Lead Time includes inspection time



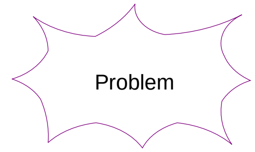
Brainstorm Improvements

As we understand the source and magnitude of waste in the value stream, we can brainstorm improvement ideas. These are typically placed as kaizen bursts on the map in the applicable area.

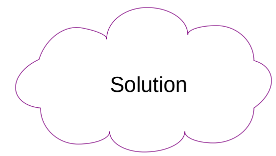


Annotating the map with improvement ideas

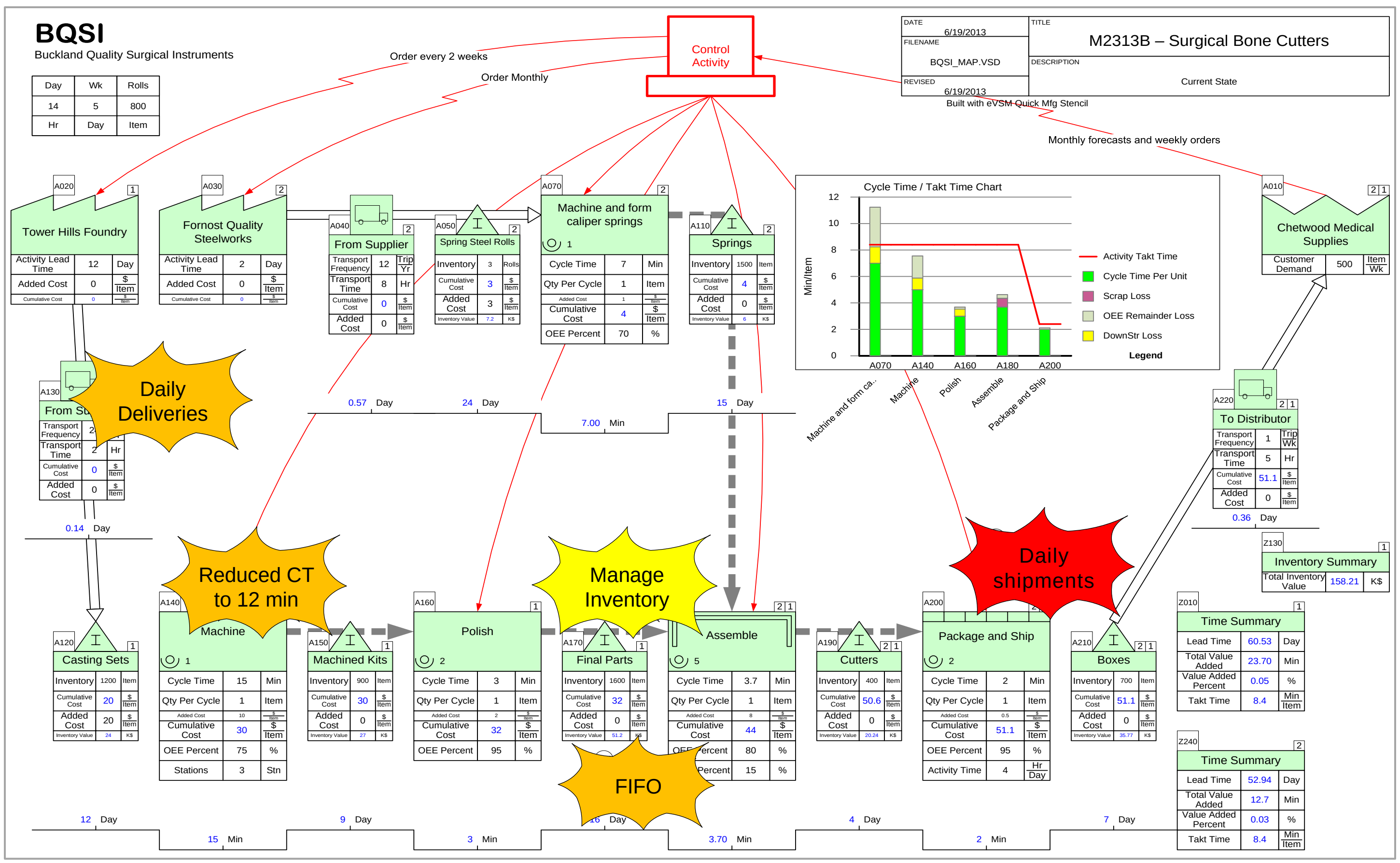
Kaizen Starbursts are used to highlight problems in the value stream



Kaizen Clouds are used to highlight potential improvements needed in the value stream

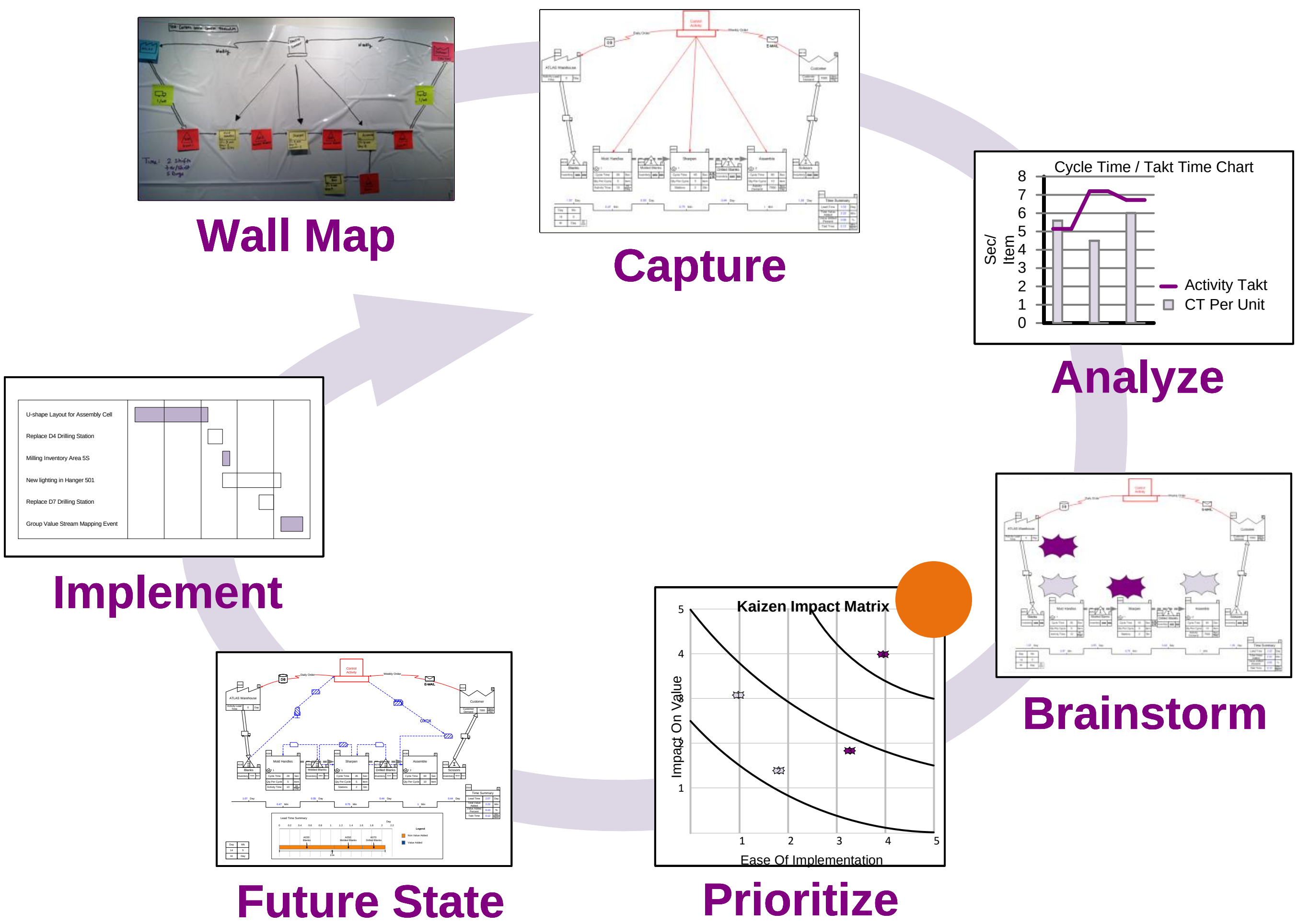


Kaizens can be color-coded also as shown below. Green could represent “Just Do It” for example.



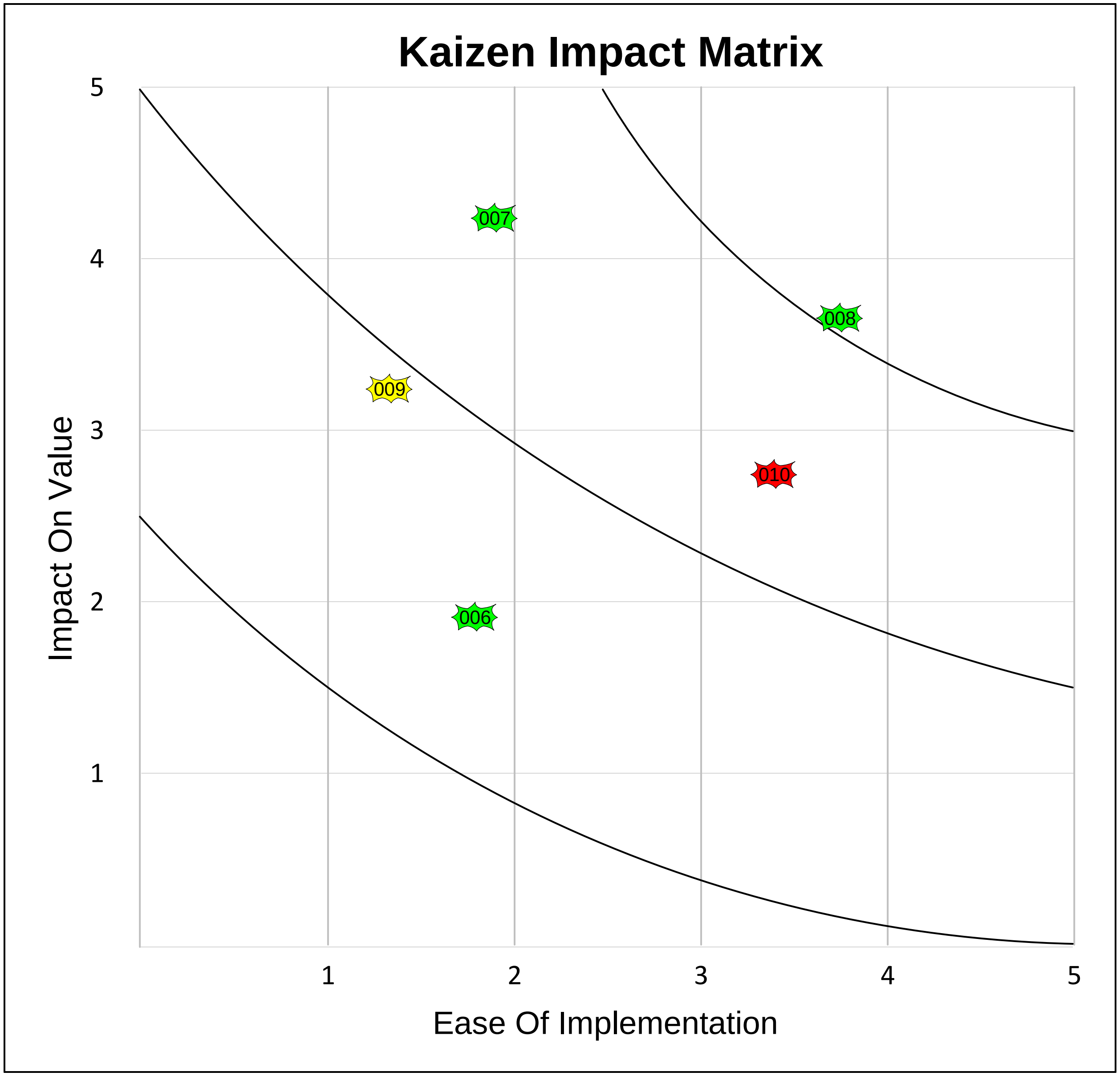
Improvement Prioritization

Next step is to select the improvements we want to implement as a next step. Color-coding the kaizen starbursts and simple prioritization matrix of impact vs. difficulty can help with this task.



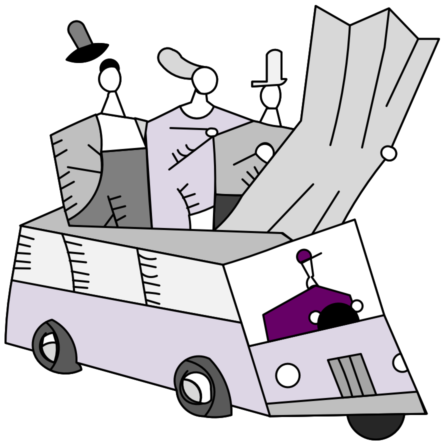
Kaizen Impact Matrix

The Impact Matrix shows different improvement ideas using our judgement of the impact of implementation versus the ease of implementation. The “best” ideas are thus on the upper right of the matrix.



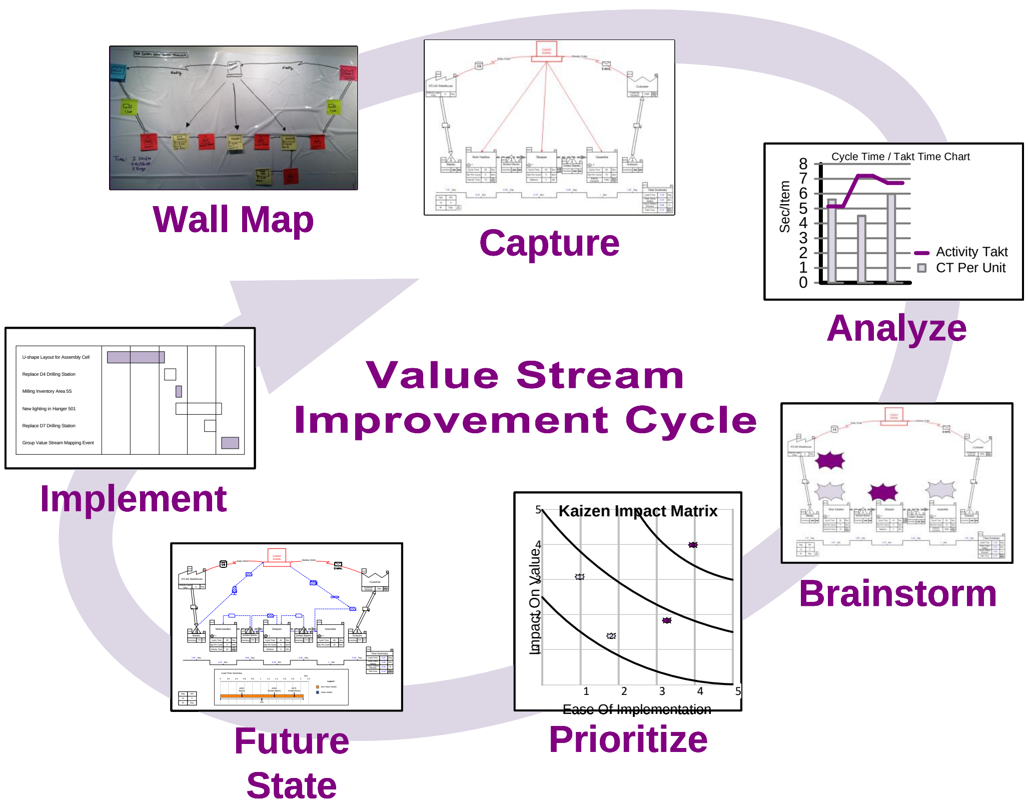
Why Value Stream Mapping

- Visualize beyond a single process
- Helps you see the “source” of waste in addition to its “magnitude”
- Provides a common language of manufacturing process
- Makes decisions apparent
- Ties in lean concepts and techniques
- Forms basis for implementation plan (“door-door”)
- Shows linkage between information & material flow
- Helps identify proper flow
- Eliminates the possible need for added Capital expenditure
- Opens CAPACITY for GROWTH



You learned that:

- Value stream maps are an early step in a lean deployment
- Maps help to understand the source of waste and it's magnitude
- Maps are created, analyzed, and modified to serve a continuous improvement cycle

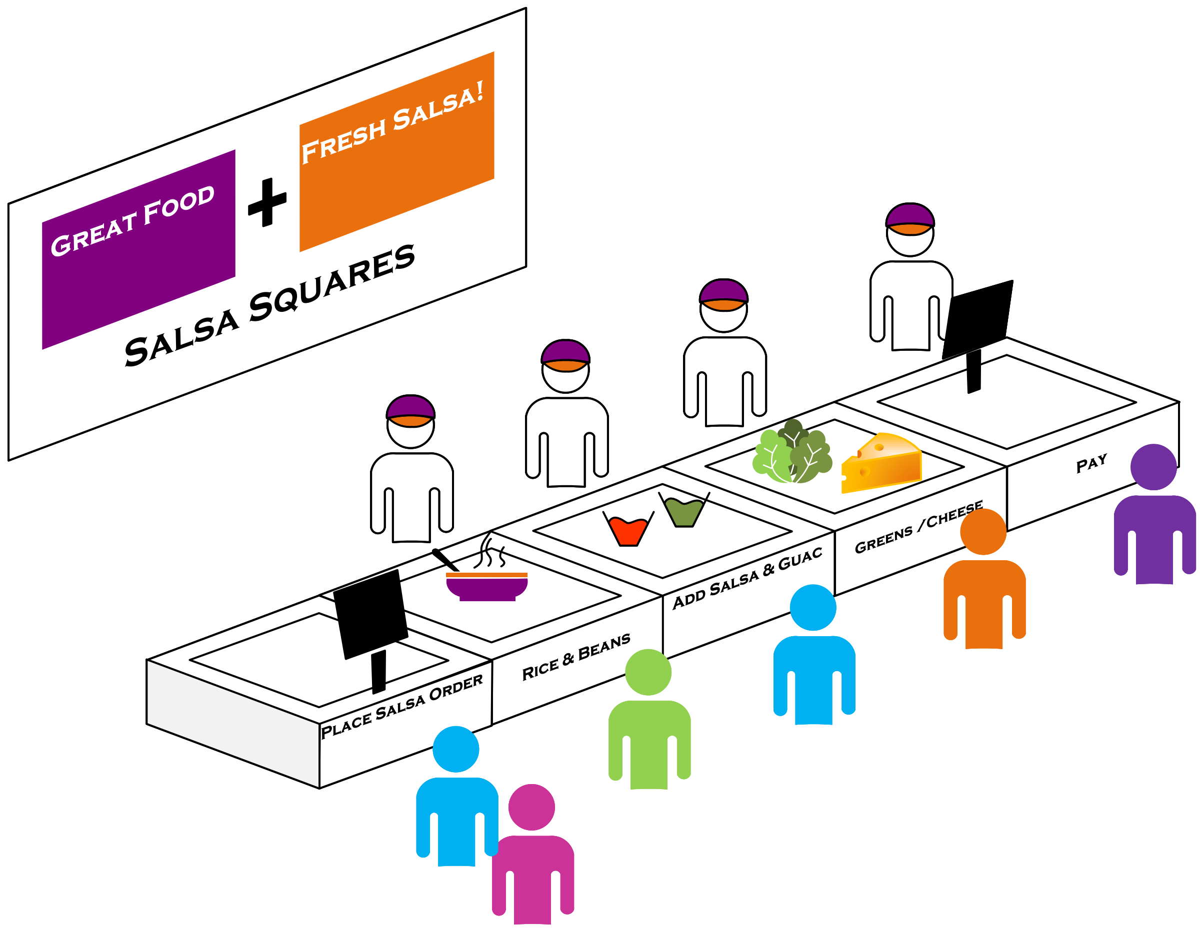


What's next:

Lean aims to remove waste in the value stream. Next you will look at the different types of waste involved.

Waste in the Value Stream

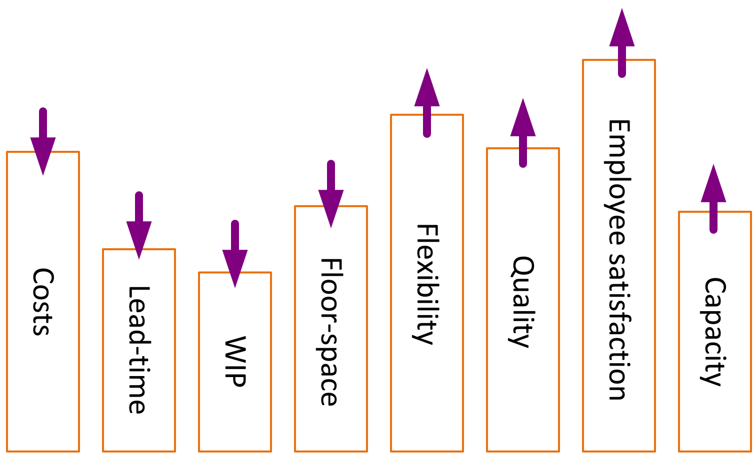
This lesson looks at the different types of waste that we are trying to remove in the value stream. We will illustrate these wastes using an example from a restaurant customer fulfillment value stream.



Waste in the Value Stream

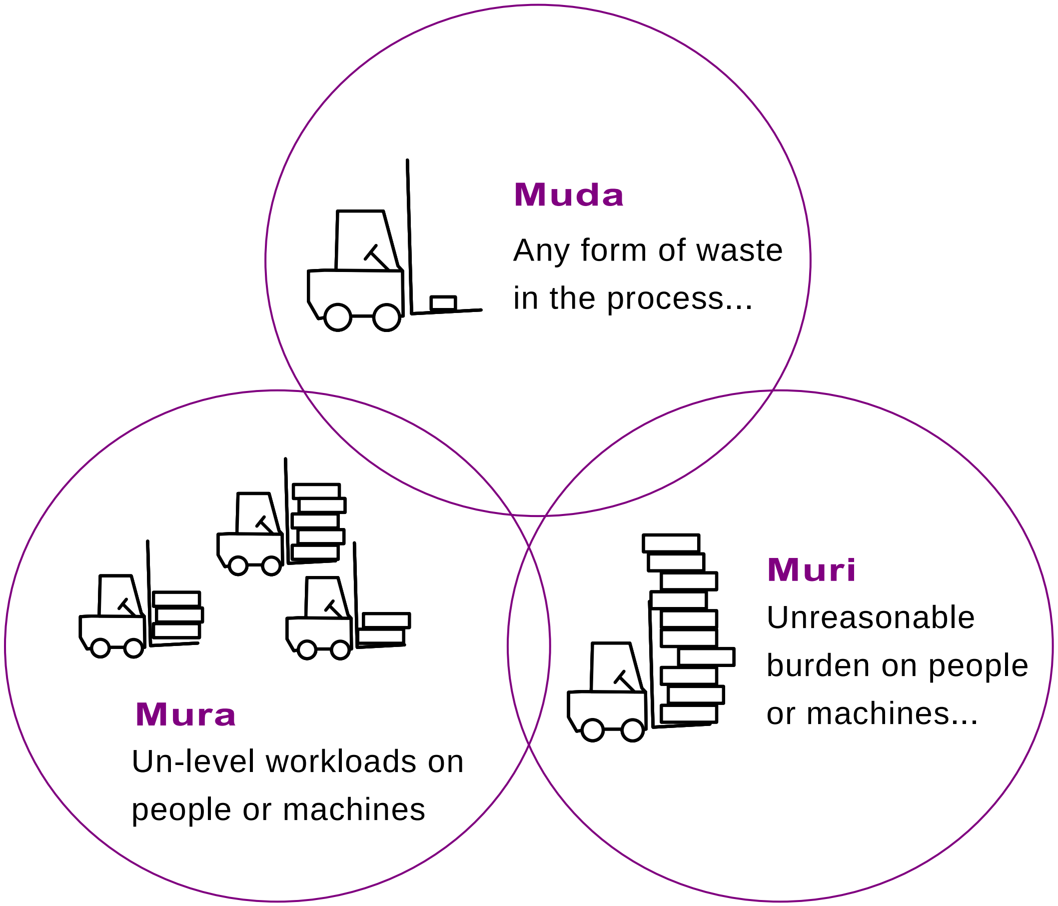
What are we trying to improve?

This simple graphic illustrates the inter-related factors we are trying to improve in a lean value stream.



Muda, Mura, Muri

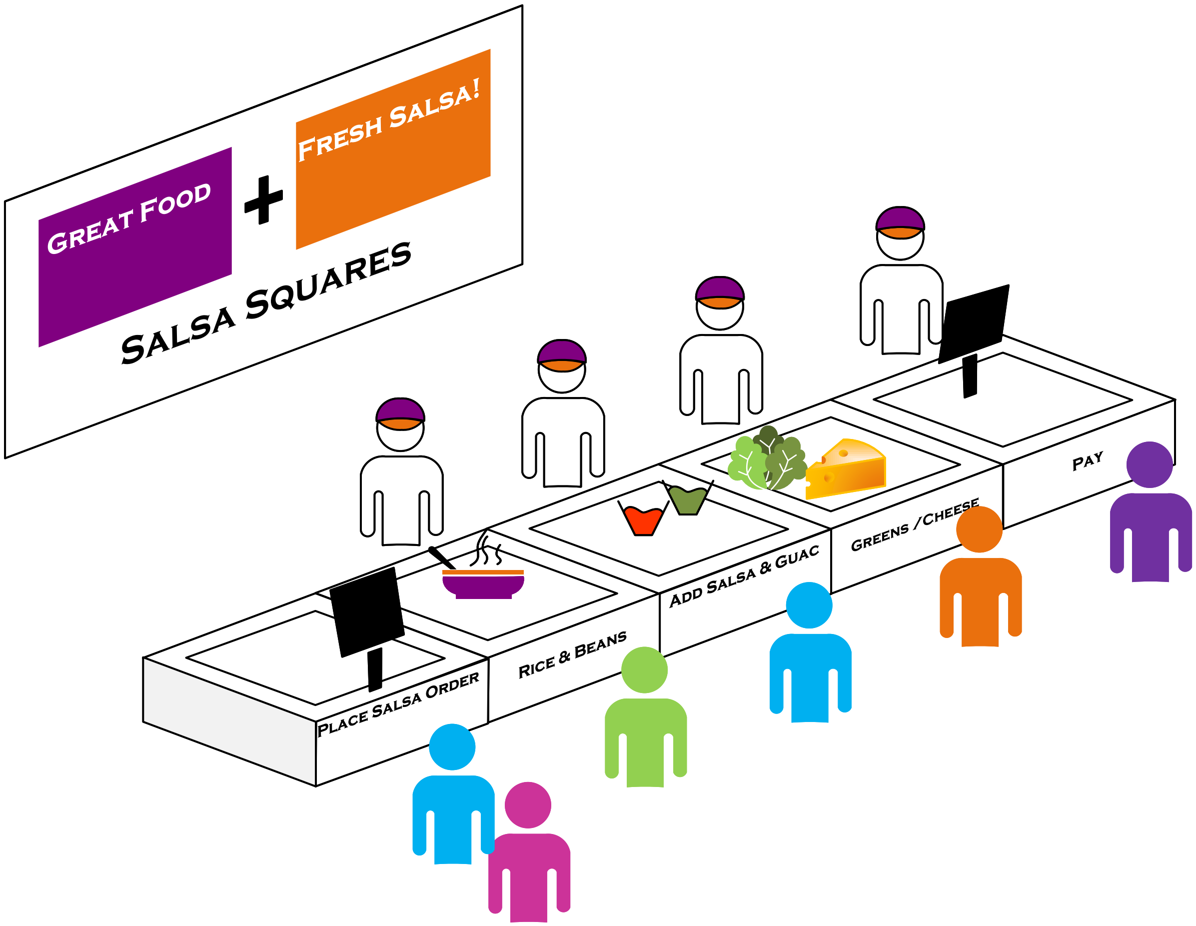
Once we map a value stream, what is it that makes it lean? There are three high level problems we are looking for. Mura, Muri, and Muda. Mura is sources of variation including unleveled workloads, Muri is unreasonable burden on people or equipment, and Muda is any form of waste in the process. We will look at the different types of Muda next.



Waste Examples at Salsa Squares

To illustrate the different types of Muda, we will look at the customer fulfillment process at a Mexican fast food restaurant.

Customers move along the counter and customize their order.



Types of Waste

Defects

Overproduction

Waiting

Not Utilizing Employees (*K,S,A)

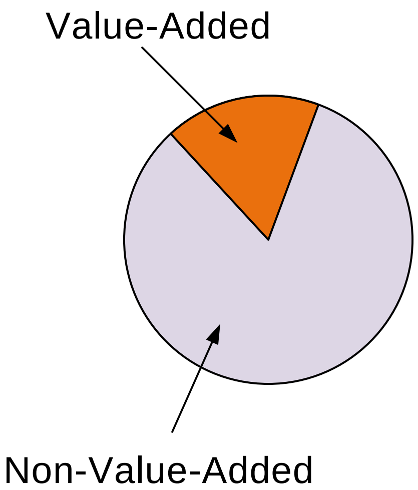
Transportation

Inventory

Motion

Excess Processing

*Knowledge, Skills, Ability



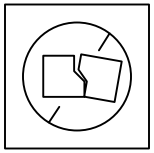
Non-Value Added and Value Added

NVA: The time spent on activities that add costs but no value to an item from the customer’s perspective.

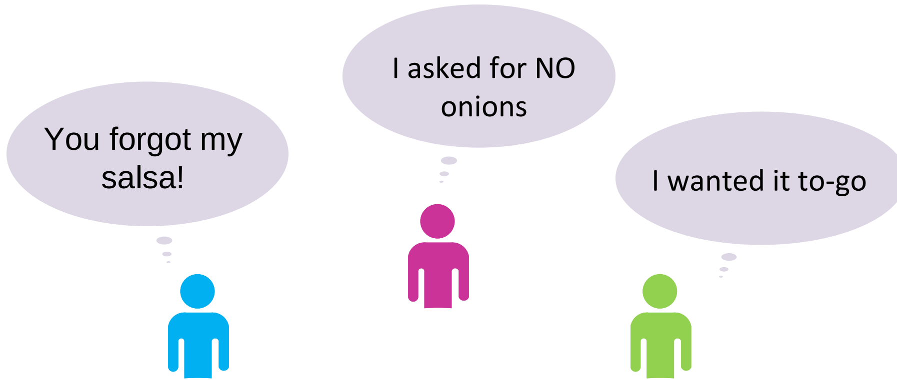
Ex: storage, inspection, rework

VA: The time of those work elements that actually transform the product in a way that the customer is willing to pay for. Usually, value-creating time is less than cycle time, which is less than production lead time.

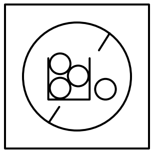
Defects



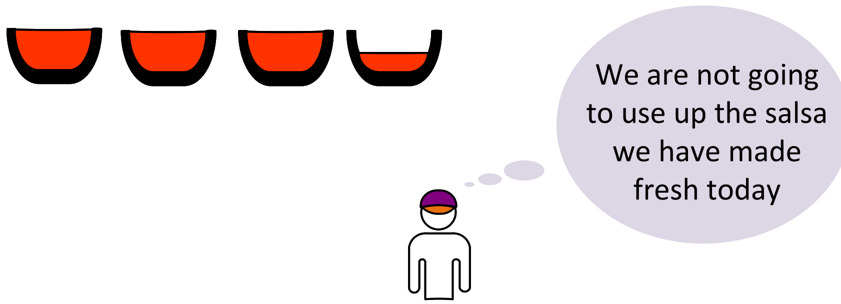
Not right the first time causes scrap or repetition of process steps



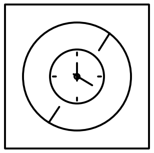
Overproduction



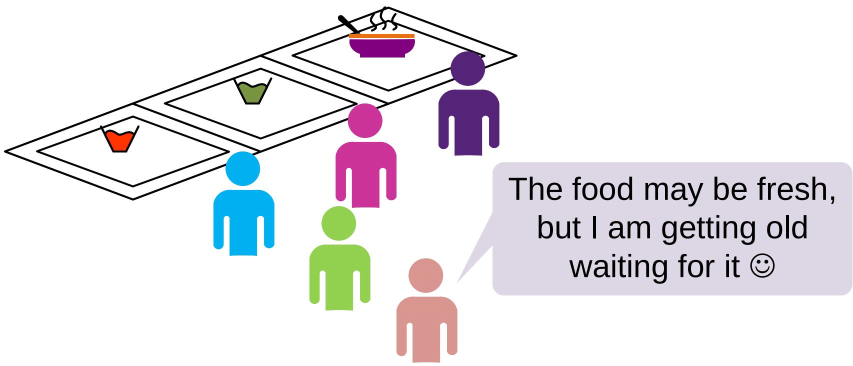
- Making more, earlier, or faster than is required by the customer
- This leads to many of the other wastes..



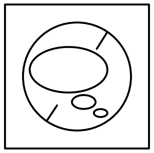
Waiting



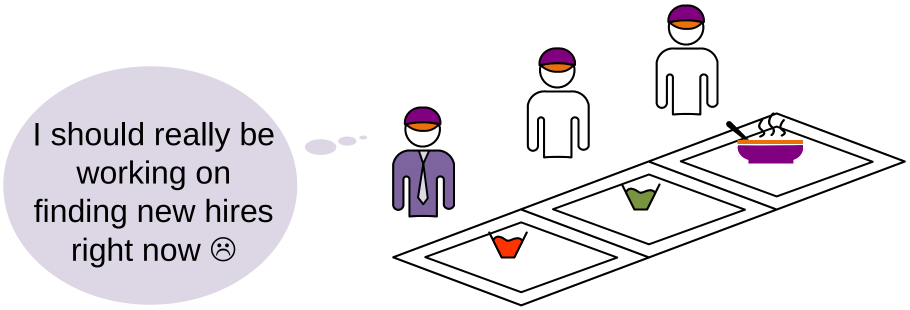
- People or parts waiting for a work cycle to be completed.



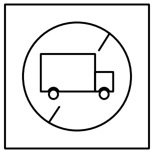
Not Utilizing Employees



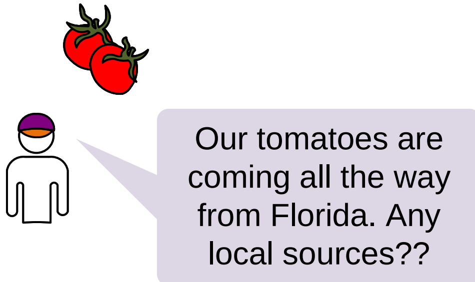
- K, S, A (Knowledge, Skills, Ability)
- Underutilization of skills



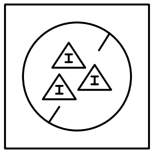
Transportation



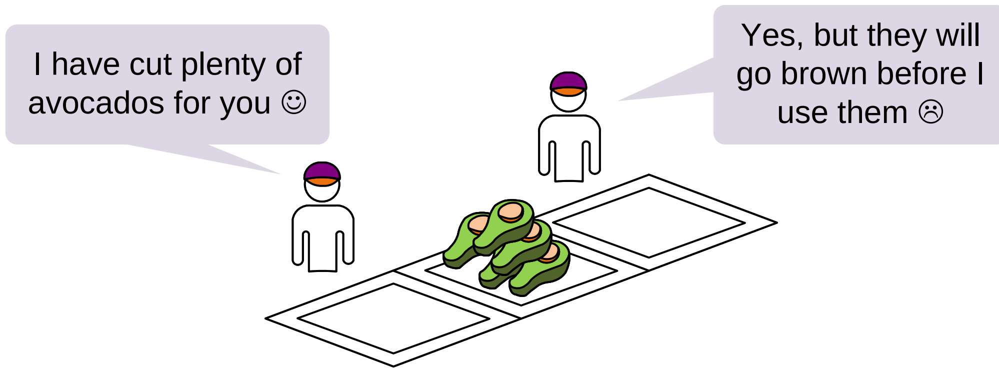
- Unnecessary conveyance of people or parts between processes



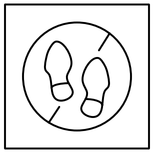
Inventory



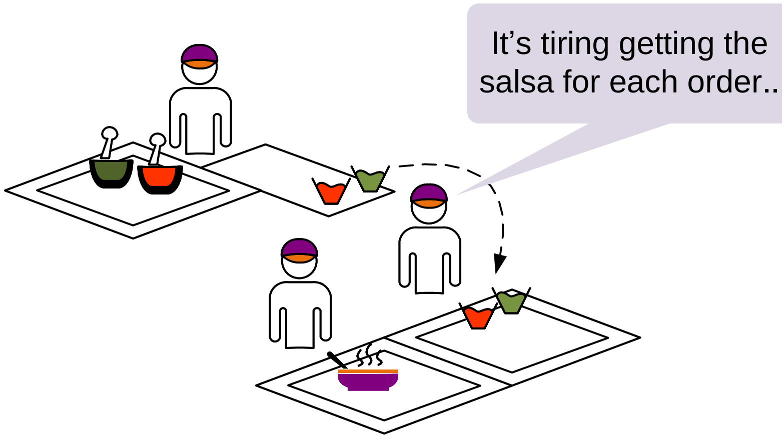
- More materials or information than is required



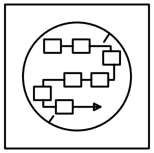
Motion



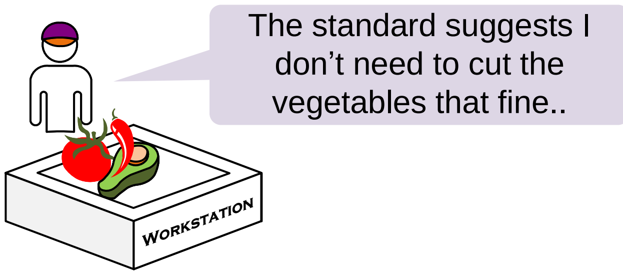
- Unnecessary movement of people, parts or equipment within a process.



Excess Processing



- Processing beyond the standard required by the customer.



Q. Overproduction is often considered the worst type of waste mainly because:

◎ It leads to many of the other wastes.

◎ It creates overburdening of staff.

◎ It creates variance in selling price in the market.

◎ It always requires overtime work.

◎ It wears out the equipment.

Q. Which of these are non-value added? Select all that apply.

☐ Material waiting to be worked upon

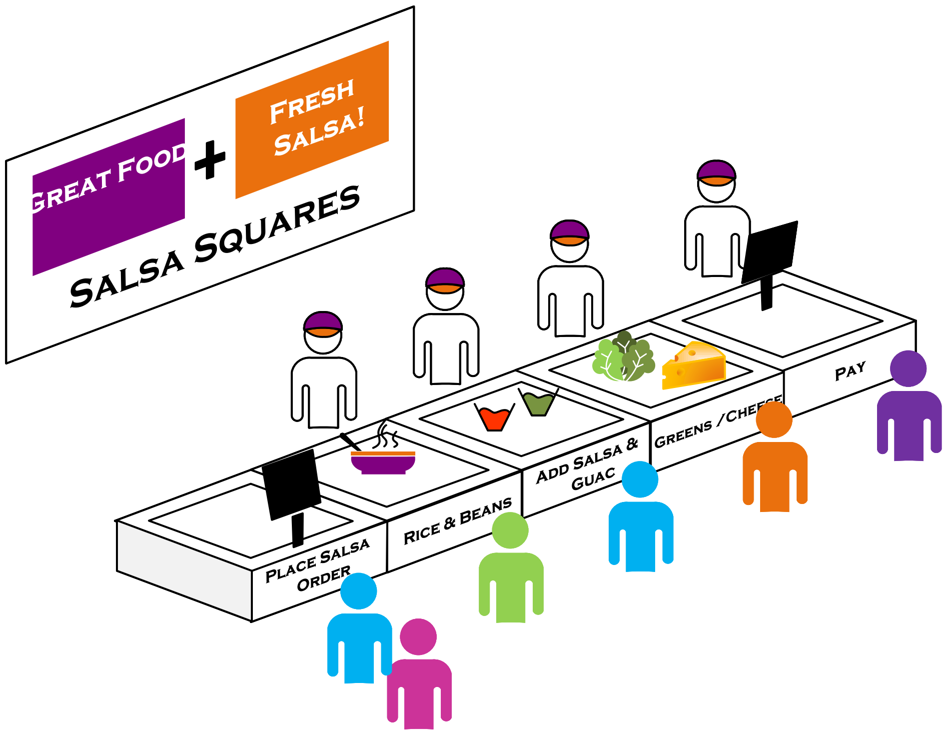
☐ People waiting to work on material

☐ Unnecessary transportation of material

☐ Over-processing beyond the customer requirement

You learned that:

- Variation (mura), over-burdening (muri), and muda are all important and must be reduced or eliminated
- There are some classic types of waste to look for in the value stream



What's next:

In this Course, you have learned about the improvement cycle using value stream maps and some of the classic wastes in the value stream. Further Courses can teach you how to quickly capture and analyze maps for improvement.

Useful Links

eVSM Toolbar Guide	evsm.com/toolbarguide
eVSM Productivity Guide	evsm.com/productivity
eVSM Blogs	evsm.com/blog
eVSM Support FAQ	evsm.com/support
Download the Latest Version	evsm.com/install